Experiment Report

1. Objective

- Familiarize with basic Docker operations, including creating, starting, stopping, and removing containers.
- Learn how to use Docker Desktop and Docker CLI to view containers, manage volumes, and control resources.
- Learn to build custom images using Dockerfile and deploy a static webpage via Nginx.
- Understand how to use Docker Compose to start multi-container applications.

2. Principle Docker is an open-source containerization platform used to package, distribute, and run applications.

- Image: A read-only template containing an application and its dependencies.
- Container: A runnable instance of an image with read-write capabilities.
- Volume: A mechanism for persisting data beyond the lifecycle of a container.
- Dockerfile: A text file defining steps to build an image.
- Docker Compose: Tool for defining and managing multi-container applications.

3. Environment

- Operating System: Windows 10/11
- Docker Version: Docker Desktop (latest)
- Terminal: PowerShell
- Browser: Any modern browser (for webpage verification)

4. Procedure

4.1 Start PostgreSQL Container

```
docker run --name db -e POSTGRES_PASSWORD=secret -d -v postgres_data:/var/lib/postgresql
/data postgres
docker exec -ti db psql -U postgres
```

4.2 Create Table and Insert Data

```
CREATE TABLE tasks (
   id SERIAL PRIMARY KEY,
   description VARCHAR(100)
);
INSERT INTO tasks (description) VALUES ('Finish work'), ('Have fun');
SELECT * FROM tasks;
\q
```

4.3 Stop and Remove Container

```
docker stop db
docker rm db
```

4.4 Use Volume for Persistent Data

docker run --name new-db -d -v postgres_data:/var/lib/postgresql/data postgres
docker exec -ti new-db psql -U postgres -c "SELECT * FROM tasks;"

4.5 Build Nginx Static Webpage Image 1. Create a public_html directory and add index.html (basic HTML page). 2. Create a Dockerfile:

```
FROM nginx:latest
COPY . /usr/share/nginx/html
EXPOSE 80
```

3. Build the image and run a container:

```
docker build -t nginx .
docker run --name web -d -p 8080:80 nginx
```

4.6 Verify Webpage Open a browser and navigate to http://localhost:8080 to check the webpage.

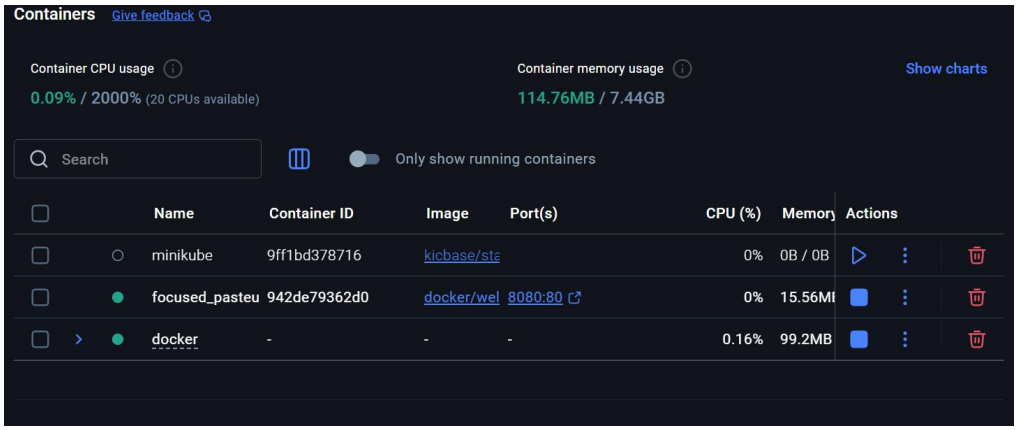
5. Results (Insert screenshots here)
- Docker Desktop container list
- PostgreSQL table creation and query
- Volume persistence verification
- Nginx webpage access

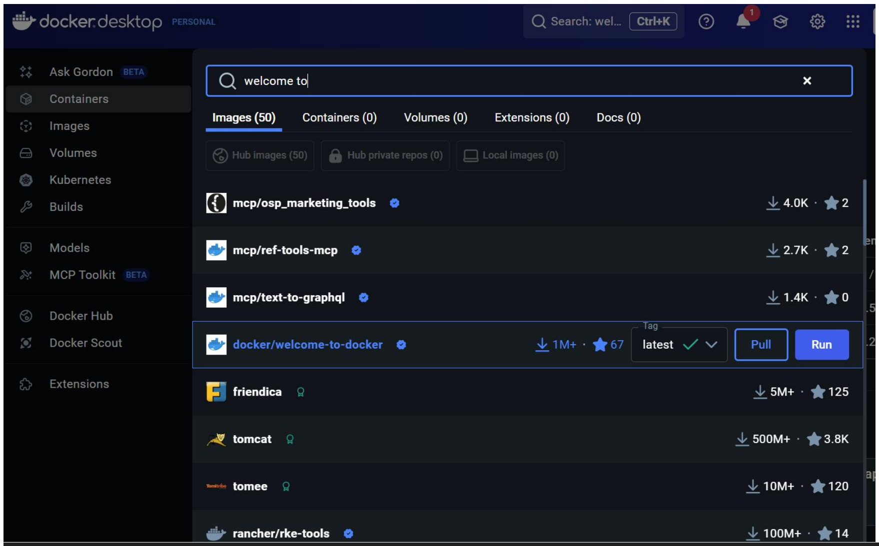
6. Analysis and Discussion
- Volumes allow data persistence even after the container is stopped or removed.
- Dockerfile makes it easy to package and deploy a static webpage quickly.
- Network issues may prevent pulling images; consider using a proxy or mirror registry.
- Docker Compose simplifies managing multi-container applications.

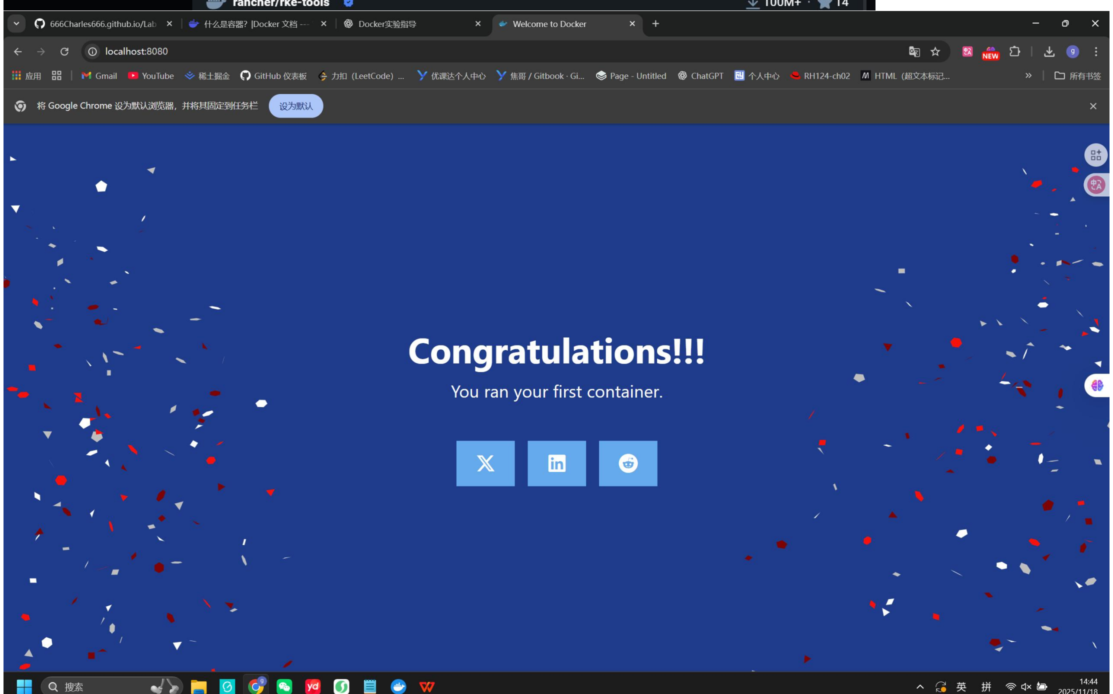
7. Conclusion
- Mastered basic Docker commands and container management.
- Understood the importance of volumes for data persistence.
- Successfully built and deployed a simple web service using Docker and Nginx.

8. Screenshot

Container



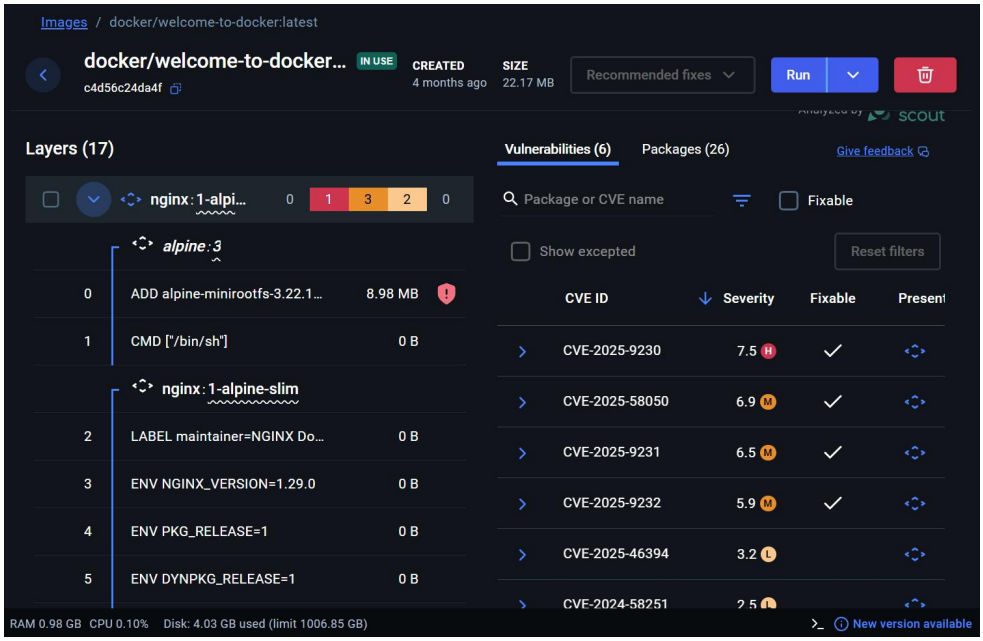




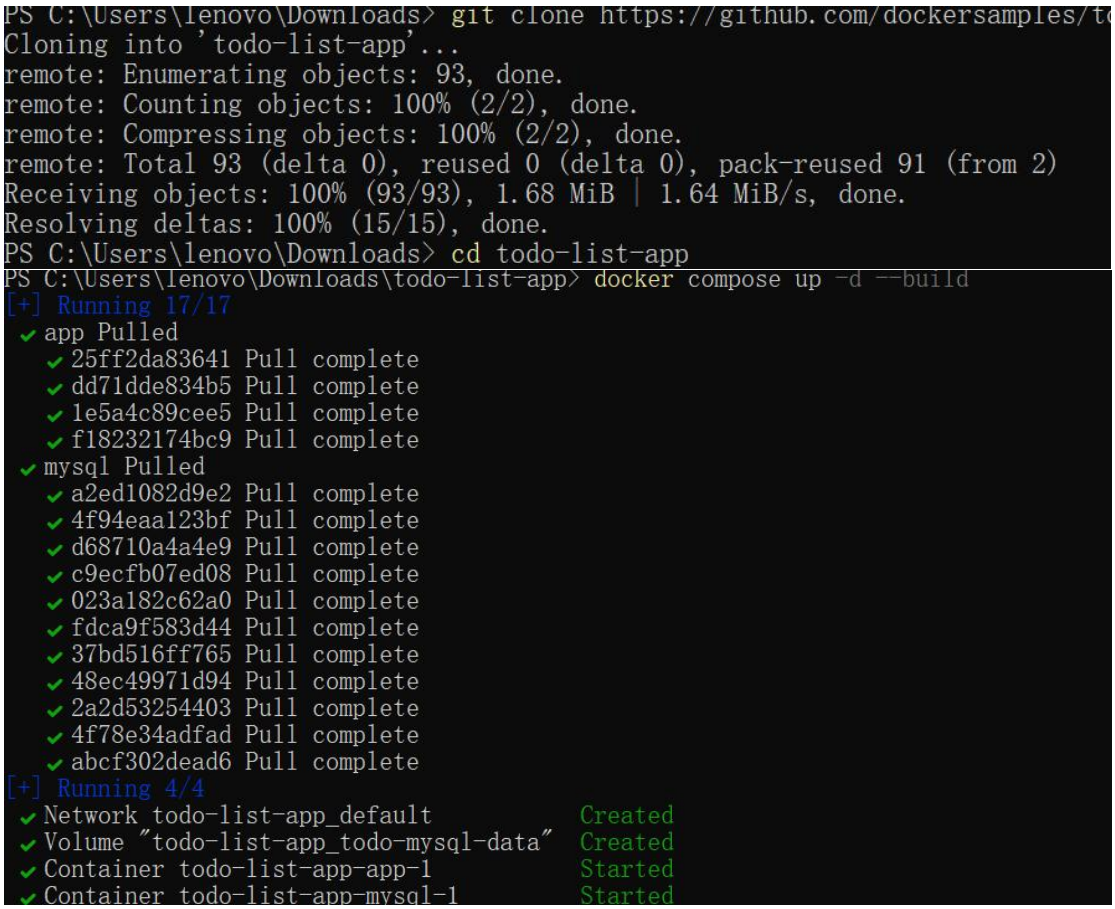

```
PS C:\WINDOWS\system32> docker run -d -p 8080:80 docker/welcome-to-docker
71a880362088c31a3daf88e05a5d6a96da705d74f17097a28418b71f516c582b
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint great_lalande (d1fc0a058298e88158bf27eb9b3667bc7a24b7a928620838dd71da82e1c81
ca6): Bind for 0.0.0.0:8080 failed: port is already allocated

Run 'docker run --help' for more information
PS C:\WINDOWS\system32> docker ps
CONTAINER ID   IMAGE                             COMMAND                  CREATED          STATUS          PORTS                                                                                NAMES
942de79362d0   docker/welcome-to-docker:latest   "/docker-entrypoint.…"   3 minutes ago    Up 3 minutes    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                                              focused_pasteur
b449c282ffc5   gitea/gitea:latest-rootless       "/usr/bin/dumb-init …"   2 months ago     Up 7 minutes    0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp, 0.0.0.0:2222->22/tcp, [::]:2222->22/tcp   gitea
```

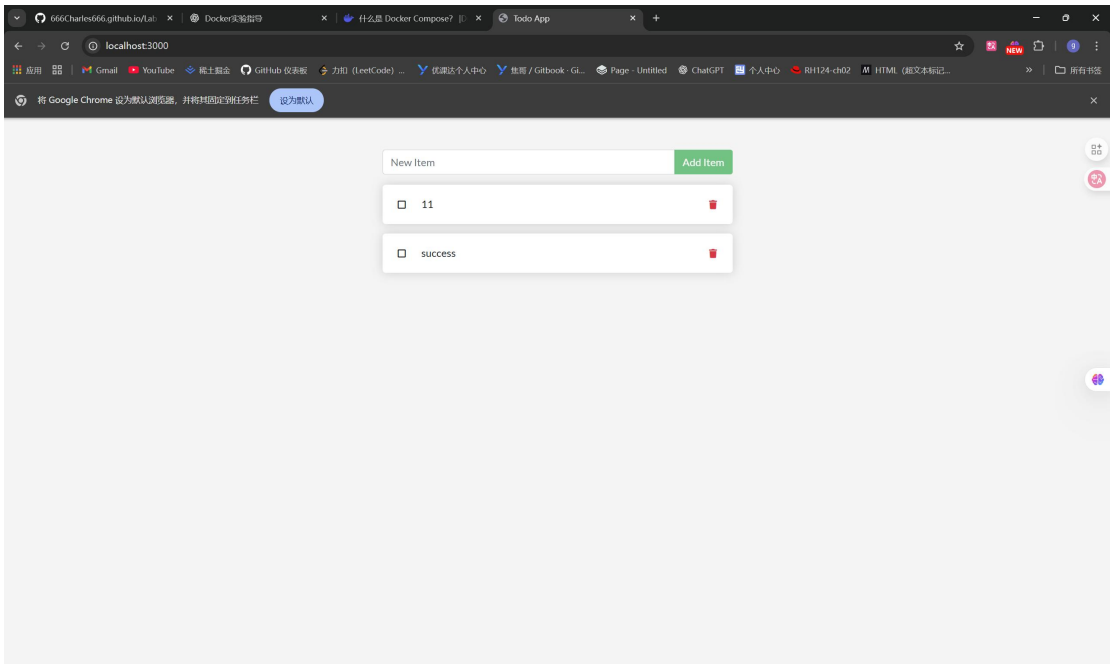
```
PS C:\WINDOWS\system32> docker stop 942de79362d0
942de79362d0
PS C:\WINDOWS\system32> docker stop b449c282ffc5
b449c282ffc5
PS C:\WINDOWS\system32> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\WINDOWS\system32> _
```

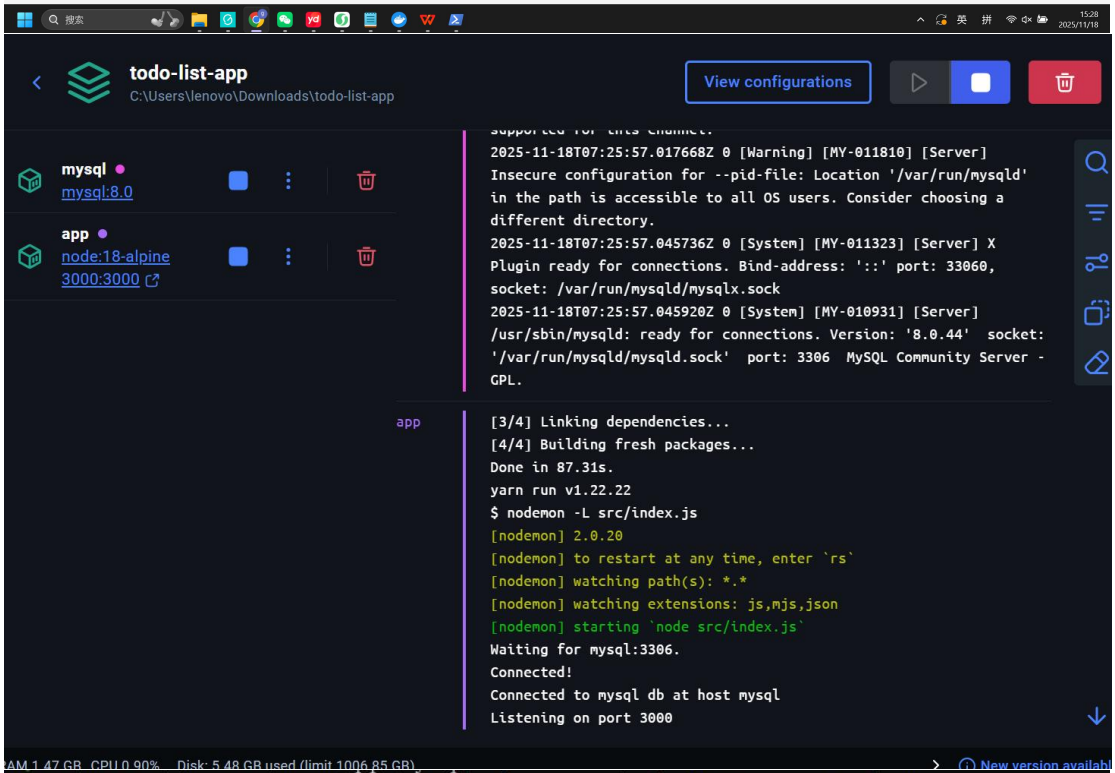
image



Docker compose

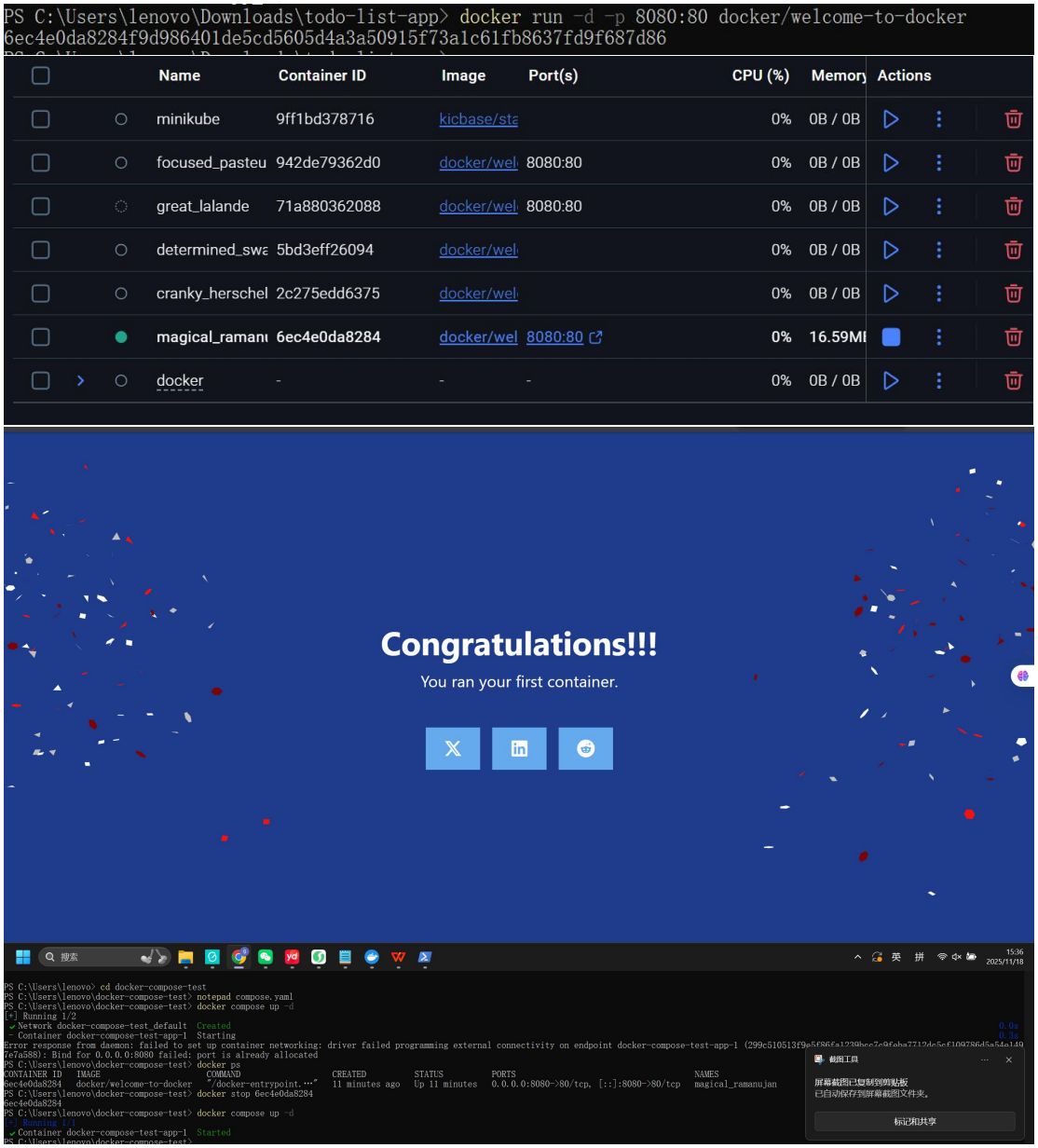






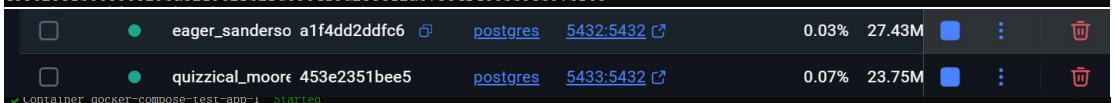

```
PS C:\Users\lenovo\Downloads\todo-list-app> docker compose down
[+] Running 3/3
 ✔ Container todo-list-app-app-1    Removed
 ✔ Container todo-list-app-mysql-1  Removed
 ✔ Network todo-list-app_default    Removed
```

Publishing and exposing ports

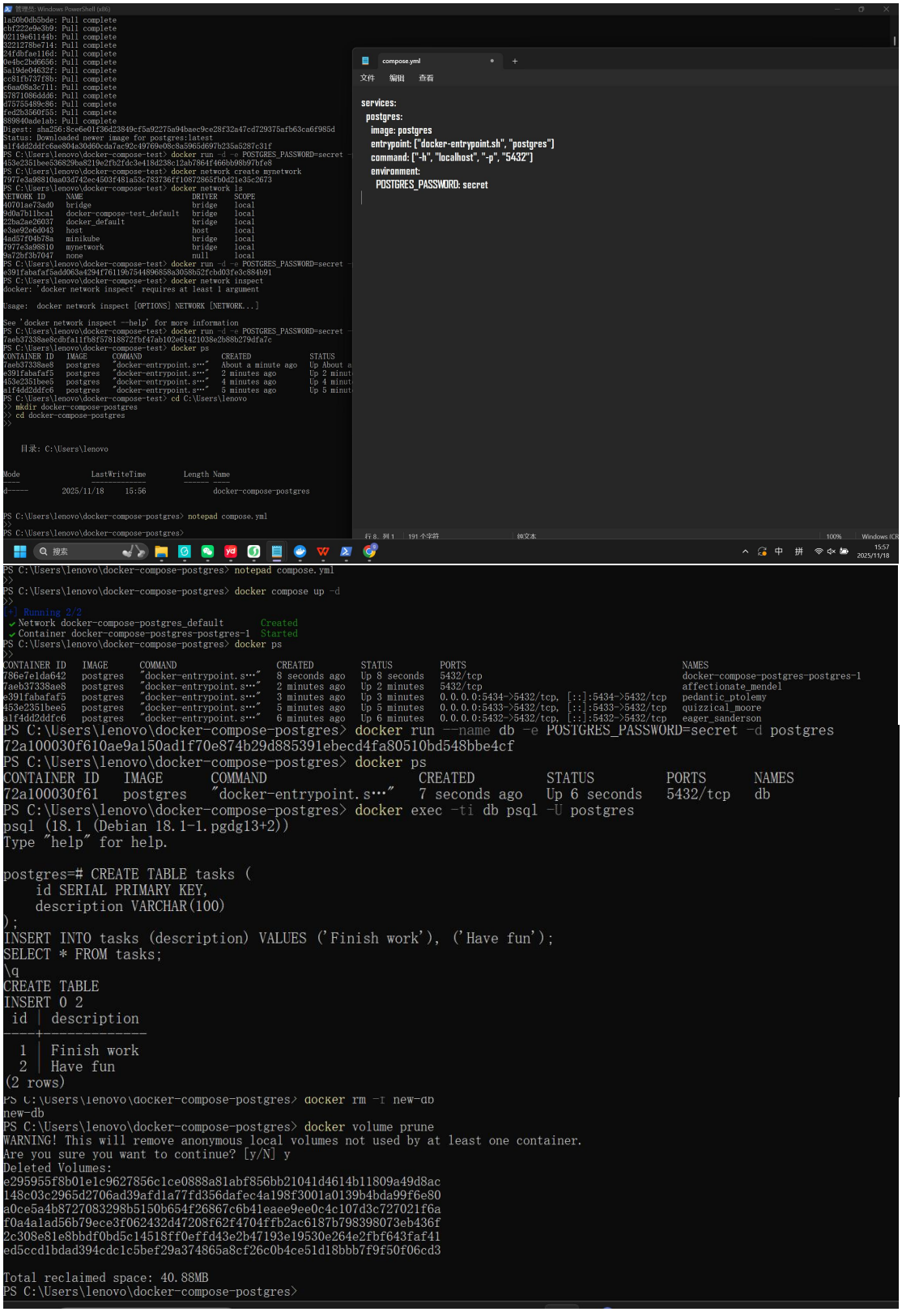


Overriding container defaults

```
PS C:\Users\lenovo\docker-compose-test> docker run -d -e POSTGRES_PASSWORD=secret -p 5432:5432 postgres
Unable to find image 'postgres:latest' locally
latest: Pulling from library/postgres
1a50b0db5bde: Pull complete
cbf222e9e3b9: Pull complete
02119e61144b: Pull complete
3221278be714: Pull complete
24fdbfae116d: Pull complete
0e4bc2bd6656: Pull complete
5a19de04632f: Pull complete
cc81fb737f8b: Pull complete
c6aa08a3c711: Pull complete
57871086ddd6: Pull complete
d75755489c86: Pull complete
fed2b3560f55: Pull complete
889840ade1ab: Pull complete
Digest: sha256:8ce6e01f36d23849cf5a92275a94baec9ce28f32a47cd729375afb63ca6f985d
Status: Downloaded newer image for postgres:latest
a1f4dd2ddfc6ae804a30d60cda7ac92c49769e08c8a5965d697b235a5287c31f
PS C:\Users\lenovo\docker-compose-test> docker run -d -e POSTGRES_PASSWORD=secret -p 5433:5432 postgres
453e2351bee536829ba8219e2fb2fdc3e418d238c12ab7864f466bb98b97bfe8
```

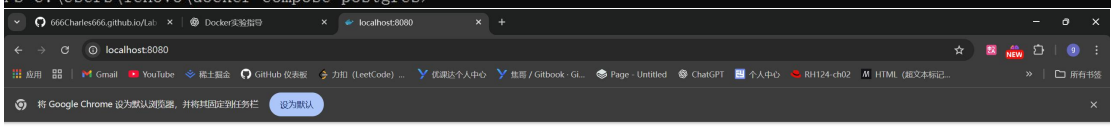



```
✔ Container docker-compose-test-app-1  Started
PS C:\Users\lenovo\docker-compose-test> docker run -d -e POSTGRES_PASSWORD=secret -p 5432:5432 postgres
Unable to find image 'postgres:latest' locally
latest: Pulling from library/postgres
1a50b0db5bde: Pull complete
cbf222e9e3b9: Pull complete
02119e61144b: Pull complete
3221278be714: Pull complete
24fdbfae116d: Pull complete
0e4bc2bd6656: Pull complete
5a19de04632f: Pull complete
cc81fb737f8b: Pull complete
c6aa08a3c711: Pull complete
57871086ddd6: Pull complete
d75755489c86: Pull complete
fed2b3560f55: Pull complete
889840ade1ab: Pull complete
Digest: sha256:8ce6e01f36d23849cf5a92275a94baec9ce28f32a47cd729375afb63ca6f985d
Status: Downloaded newer image for postgres:latest
a1f4dd2ddfc6ae804a30d60cda7ac92c49769e08c8a5965d697b235a5287c31f
PS C:\Users\lenovo\docker-compose-test> docker run -d -e POSTGRES_PASSWORD=secret -p 5433:5432 postgres
453e2351bee536829ba8219e2fb2fdc3e418d238c12ab7864f466bb98b97bfe8
PS C:\Users\lenovo\docker-compose-test> docker network create mynetwork
7977e3a98810aa03d742ec4503f481a53c783736ff10872865fb0d21e35c2673
PS C:\Users\lenovo\docker-compose-test> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
40701ae73ad0   bridge                        bridge    local
9d0a7b11bca1   docker-compose-test_default   bridge    local
22ba2ae26037   docker_default                bridge    local
e3ae92e6d043   host                          host      local
4ad57f04b78a   minikube                      bridge    local
7977e3a98810   mynetwork                     bridge    local
9a72bf3b7047   none                          null      local
PS C:\Users\lenovo\docker-compose-test> docker run -d -e POSTGRES_PASSWORD=secret -p 5434:5432 --network mynetwork postgres
e391fabafaf5add063a4294f76119b7544896858a3058b52fcbd03fe3c884b91
PS C:\Users\lenovo\docker-compose-test> docker network inspect
docker: 'docker network inspect' requires at least 1 argument

Usage:  docker network inspect [OPTIONS] NETWORK [NETWORK...]

See 'docker network inspect --help' for more information
PS C:\Users\lenovo\docker-compose-test>
```

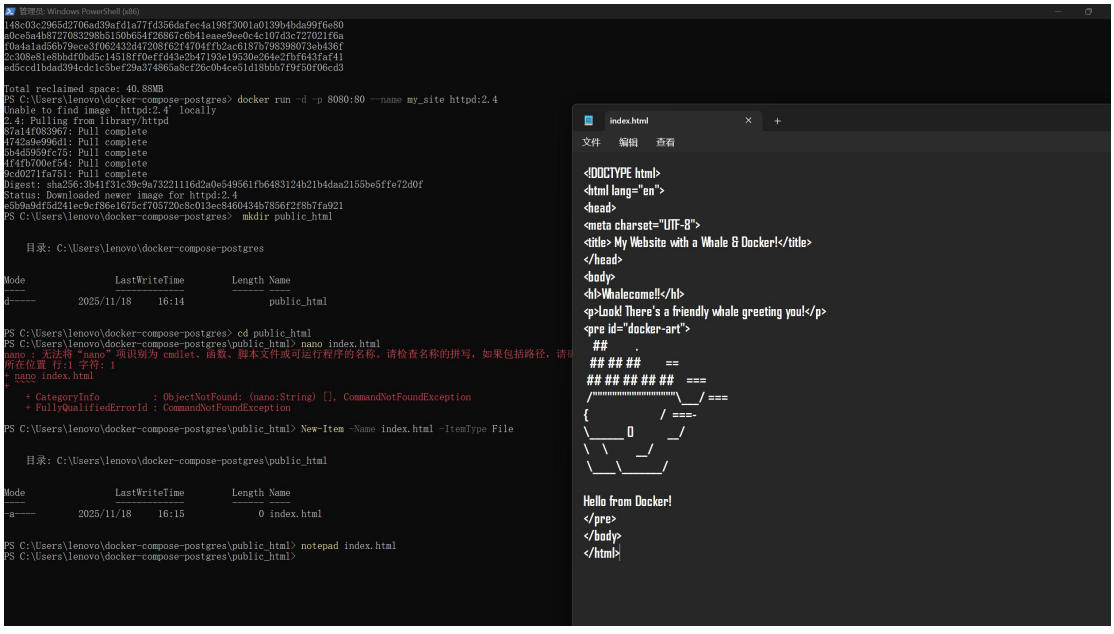


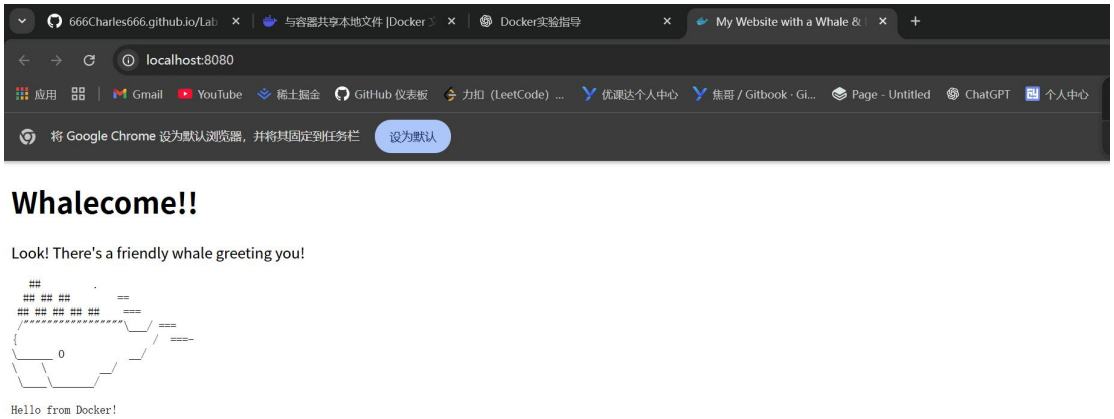
Sharing local files with containers

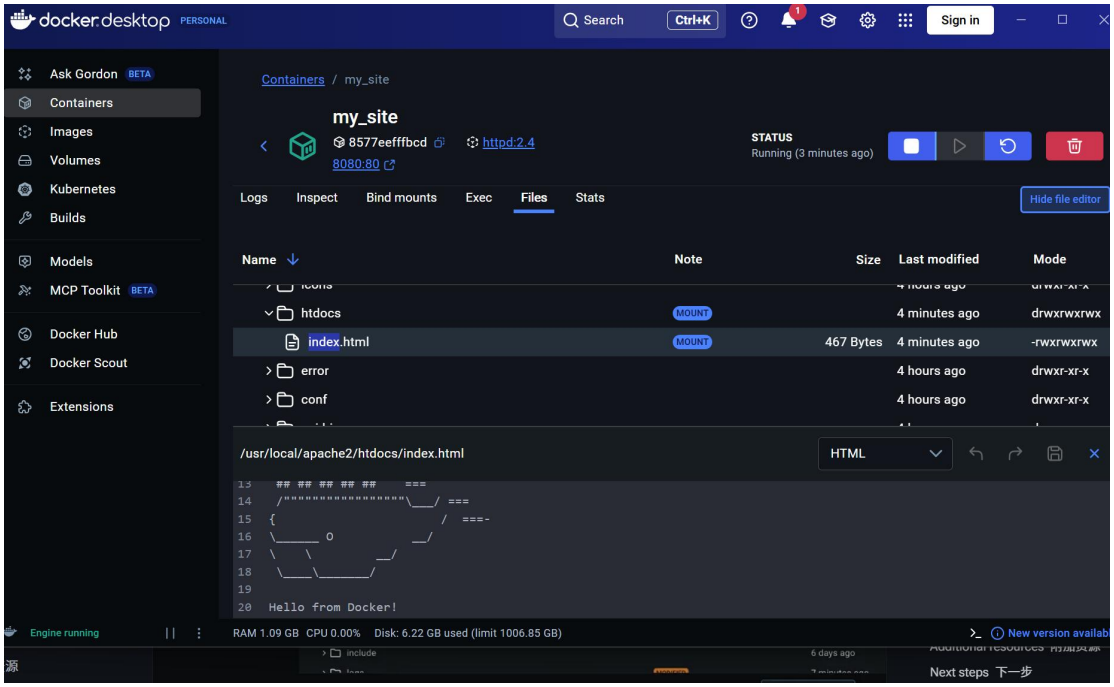
```
PS C:\Users\lenovo\docker-compose-postgres> docker run -d -p 8080:80 --name my_site httpd:2.4
Unable to find image 'httpd:2.4' locally
2.4: Pulling from library/httpd
87a14f083967: Pull complete
4742a9e996d1: Pull complete
5b4d5959fc75: Pull complete
4f4fb700ef54: Pull complete
9cd0271fa751: Pull complete
Digest: sha256:3b41f31c39c9a73221116d2a0e549561fb6483124b21b4daa2155be5ffe72d0f
Status: Downloaded newer image for httpd:2.4
e5b9a9df5d241ec9cf86e1675cf705720c8c013ec8460434b7856f2f8b7fa921
PS C:\Users\lenovo\docker-compose-postgres>
```



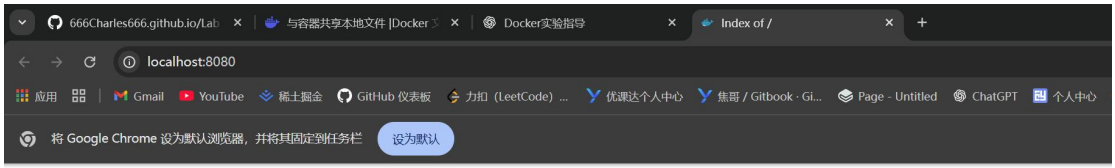
It works!



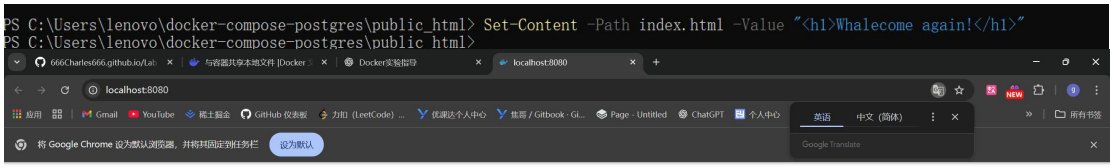

666Charles666.github.io/Lab
与容器共享本地文件 |Docker
Docker实验指导
My Website with a Whale &
localhost:8080
应用
Gmail
YouTube
稀土掘金
GitHub 仪表板
力扣（LeetCode）...
优课达个人中心
焦哥 / Gitbook · Gi...
Page - Untitled
ChatGPT
个人中心
将 Google Chrome 设为默认浏览器，并将其固定到任务栏
设为默认
Whalecome!!
Look! There's a friendly whale greeting you!
Hello from Docker!


docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Ask Gordon
BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Docker Scout
Extensions
Containers / my_site
my_site
8577eefffbcd
httpd:2.4
8080:80
STATUS
Running (3 minutes ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
Hide file editor
Name
Note
Size
Last modified
Mode
htdocs
MOUNT
4 minutes ago
drwxrwxrwx
index.html
MOUNT
467 Bytes
4 minutes ago
-rwxrwxrwx
error
4 hours ago
drwxr-xr-x
conf
4 hours ago
drwxr-xr-x
/usr/local/apache2/htdocs/index.html
HTML
Hello from Docker!
Engine running
RAM 1.09 GB CPU 0.00%
Disk: 6.22 GB used (limit 1006.85 GB)
New version availabl
Next steps 下一步

When delete

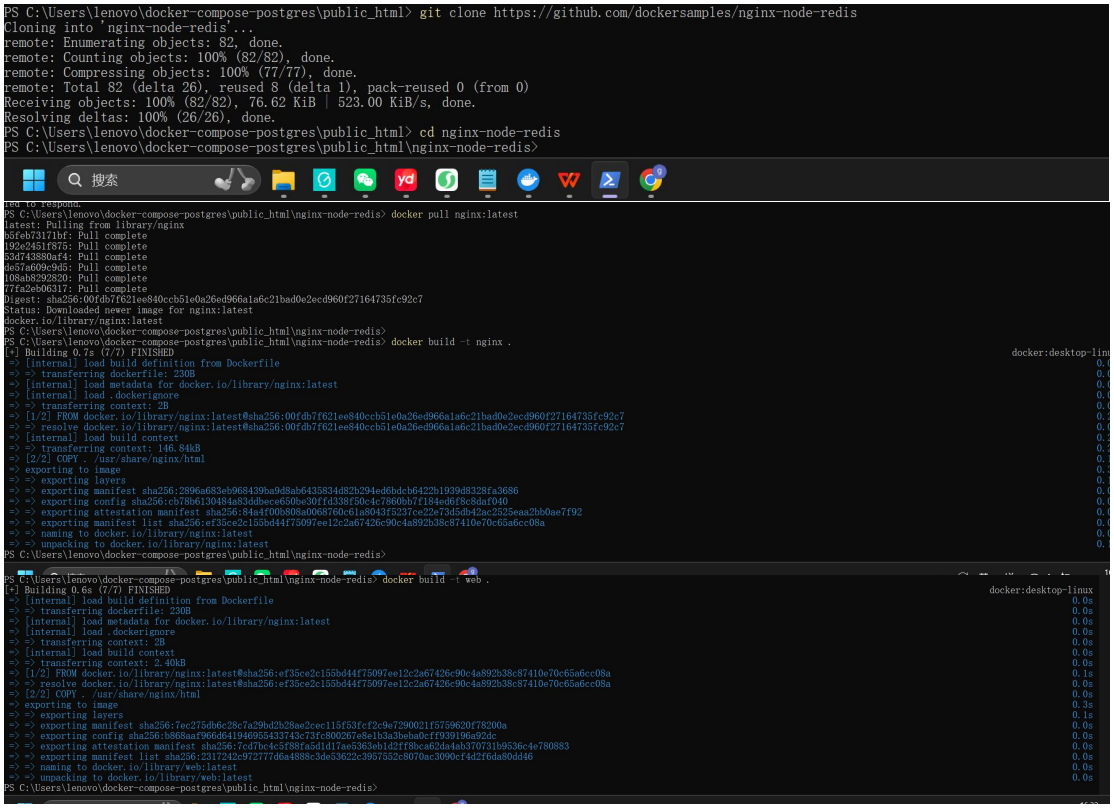


Index of /



Whalecome again!

Multi-container applications



```
PS C:\Users\lenovo\docker-compose-postgres\public_html\nginx-node-redis> docker network create sample-app
7fb52439e1c454831057a095e3ada2413fd90cf0b84ce7bcee6ed2e6919e7538
PS C:\Users\lenovo\docker-compose-postgres\public_html\nginx-node-redis> docker run -d  --name redis --network sample-app --network-alias redis redis
Unable to find image 'redis:latest' locally
latest: Pulling from library/redis
ae59e22f9916: Pull complete
5be4bf37d661: Pull complete
2e0d85c0940e: Pull complete
4f4fb700ef54: Pull complete
5773a6174dac: Pull complete
c87cd8c277cf: Pull complete
3e44f01296e3: Pull complete
Digest: sha256:49887660cd57f34b2b3bfb73ae135e2265320c48c3d76a9294e9851265e140d6
Status: Downloaded newer image for redis:latest
ef9e33158dc9cbf3406e020fdf9d4a13902638567debf2e4c79751cf4d2a3fd4
PS C:\Users\lenovo\docker-compose-postgres\public_html\nginx-node-redis> docker run -d --name web1 -h web1 --network sample-app --network-alias web1 web
459096270ee93bfd4b1557ae6fce9154f741ae9720a0a91dbfff9479745754e5
PS C:\Users\lenovo\docker-compose-postgres\public_html\nginx-node-redis> docker run -d --name web2 -h web2 --network sample-app --network-alias web2 web
6f9d20511280b029db24f2f30cbdc360fb7bae5d358ff352bdf7322d7ef8799b
PS C:\Users\lenovo\docker-compose-postgres\public_html\nginx-node-redis> docker run -d --name nginx --network sample-app  -p 80:80 nginx
314cf39834fffa4454e3d0adabe1b7237b60a8a39be140db9fbc215cde7cee1d
PS C:\Users\lenovo\docker-compose-postgres\public_html\nginx-node-redis> docker ps
CONTAINER ID   IMAGE        COMMAND                  CREATED          STATUS          PORTS                                   NAMES
314cf39834ff   nginx        "/docker-entrypoint.…"   8 seconds ago    Up 8 seconds    0.0.0.0:80->80/tcp, [::]:80->80/tcp     nginx
6f9d20511280   web          "/docker-entrypoint.…"   17 seconds ago   Up 17 seconds   80/tcp                                  web2
459096270ee9   web          "/docker-entrypoint.…"   27 seconds ago   Up 27 seconds   80/tcp                                  web1
ef9e33158dc9   redis        "docker-entrypoint.s…"   39 seconds ago   Up 39 seconds   6379/tcp                                redis
8577eefffbcd   httpd:2.4    "httpd-foreground"       16 minutes ago   Up 16 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp my_site
PS C:\Users\lenovo\docker-compose-postgres\public_html\nginx-node-redis>
```